PUJA MONDAL

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LinkedIn



OBJECTIVE

- AI Application Develper and ML engineer seeks opportunities to solve real world problems.
- Having a technical background and business analytical mind help to collaborate with team in order to reach defined goal.
- Experience in full life cycle development of chat bots and object detection and segmentation of computer vision in real time enterprise environment.

TECHNICAL SKILLS

Languages : Python , Java, C

Technologies & Tools : Computer Vision, Prompt Engineering, NLP, GenAI, LLM, RAG, Flask, FastAPI, Tensoflow, Pytorch,Pandas, Matplotlib, HuggingFace, Deep Learning, Data Annotation(Labelme, Labellimg, Roboflow), MySQL, SQLite, SQLAlchemy(ORM)

WORK EXPERIENCE

Organization: Nxerra Technologies, Kolkata JUNE 2025 – PRESENT

AI/ML Engineer

- **AI in Agriculture**
  -- Prepare and optimize dataset using data engineering; Implement ML algorithm and develop Computer Vision model for Crop Management (Features- crop disease detection, weed detection, crop recognition, grading by quality, yield prediction)
  -- Developed ML model for Soil Management (Features- Insect Detection, Fertilizer Recommendation)

  Contributed to building end-to-end AI pipelines (Data preprocessing, Model training, Evaluation, Inference)

Organization: Actovision IT Solutions Pvt Ltd, Kolkata JUNE 2024 – MAY 2025

Machine Learning Engineer

- Responsible for the technical implementation of the chatbot api integration and testing through postman.
- Backend API integration through Flask and FastAPI and communicate with MySQL , SQLite Databases.
- Responsible for real time conversational bot development starting from development phase getting requirement, bot development, testing.
- Giving consultation and Develop AI Builder content from scratch using OpenAI API and Prompt Engineering Technique.
- Develop text to image generation chatbots using Dall-E from OpenAI API.
- Work on both with voice and text-based chatbots using OpenAI, Gemini, Llama model.
- Giving consultations and recommendations on chatbot development and implement from scratch (Student chatbot, financial assistant) using Prompt Engineering.
- Develop different kind of bot (Funbot, Contentcrafter, Health helper, Recipe radar, Career coach, Language learner, Tech tutor, Marketing assistant, Email assistant)
- Fine Tuning LLM on the Indian Law Dataset using Llama , LORA, Qlora, PEFT technique as well as Fine Tune LLM using Mistral model.

Organization: MavenAI Technologies Pvt Ltd, Chennai, India AUGUST 2023 – MAY 2024

Machine Learning and Computer vision intern

- Worked on building machine learning models for Detection of Desert Tree on Drone Images & Videos
- Model Development for Detection of Crop Damages in Drone Images & Videos using YOLOV8 algorithm
- Model Development for Oil Spill Detection.
- Model Development for Wild Life Animal Detection(in real-time)in Drone Videos.
- Model Development to Detect the Defect of Electric Tower.

- Model Development for Road Damage Detection.
- Model Development for Railway Track Defect Detection.
- Grounding DINO & SAM for Object Detection and Segmentation
- Hands on experience on Image annotation, Video annotation(Road Object extraction,Indidan street, Car Parts, Building Parts,workers, wild life animal, solar panel)
- NLP-based Python Code Generation for Efficient Data Analysis with Pandas & Langchain Agent On Single/Multiple databases.
- Car Accident Rule Application Chatbot with Rule-based Decision Making
- T-Rex2: Towards Generic Object Detection via Text-Visual Prompt Synergy

EDUCATION

Bachelor of Technology, Information Technology — July 2020 – July 2024
Government College of Engineering and Leather Technology,
North 24 Parganas, India (CGPA-9.06/10)

Higher Secondary School, Science Stream — June 2017 – Mar 2019
Sarsuna High School(H.S), Kolkata, India (Score –85.4%)

Secondary School Education — June 2010 – Mar 2017
Sarsuna Girls High School(H.S), Kolkata, India (Score – 85.14%)

TRAINING AND CERTIFICATION

- Certification of Industrial Training on Python Programming from A Technologies .
- Certificate of completion TCS ESG virtual experience program.
- Machine Learning and Deep Learning specialization on Coursera